

Erdős Problem #126

EP-126

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $f(n)$ be maximal such that if $A \subseteq \mathbb{N}$ has $|A| = n$ then $\prod_{a \neq b \in A} (a + b)$ has at least $f(n)$ distinct prime factors. Is it true that $f(n)/\log n \rightarrow \infty$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>